

Bhuvesh Kumar

Senior Research Scientist · Ph.D. in Computer Science

✉ kumarbhuvesh@gmail.com | 🌐 www.bhuveshkumar.com | 💻 bhuvesh-kumar | 📄 Google Scholar

Experience

Snap Inc.

SENIOR RESEARCH SCIENTIST

Bellevue, WA

Sept. 2024 - Present

- Drive research and development in LLM/VLM-based generative recommendation, multimodal Semantic ID representations and content understanding, and long user-sequence modeling. Co-created Snap's internal Python/PyTorch research library and now serve as its sole maintainer and technical lead; it powers Semantic ID and generative-recommendation use cases throughout Snap and is actively used by every research scientist on the team and 20+ MLEs.
- Developed multimodal Semantic ID content representations now used as features in 10+ production models across Content, Ads, Lens, and Search and as the target representation for Snap's generative retrieval models.
- Advanced Semantic-ID-based generative recommendation by designing and developing ID-to-text alignment strategies, user-sequence modeling objectives, implicit and explicit reasoning methods, reinforcement-learning post-training with verifiable rewards and learned reward models, and fast TensorRT-LLM inference. This work contributed to multiple production launches.
- Developed cross-domain universal user representations and long-sequence models with sharded embedding tables using TorchRec and NVIDIA DynamicEmb. The resulting Universal User Embedding Model serves as a foundational cross-domain user representation, with its embeddings used by 15+ personalization models at Snap.
- Co-created GRID, the open-source subset of this platform; GRID has 700+ GitHub stars, and its accompanying paper received the CIKM 2025 Best Paper Award.

TikTok

MACHINE LEARNING SCIENTIST

Bellevue, WA

Jan. 2023 - Sept. 2024

- Developed large-scale sequential user models and personalization algorithms for TikTok U.S. Core Recommendation.
- Introduced and owned a new final re-ranking stage that moved TikTok U.S. recommendation from point-wise item scoring to list/session-level Transformer sequence modeling, increasing total watch time by 2.4%.
- Designed a personalized mixing stage for organic, cold-start, and commercial content; coordinated with horizontal teams to preserve cold-start and commercial-content distribution goals while optimizing long-term user retention.
- Led the migration of TikTok U.S. Friends, Hashtag, and Social recommendation services from a single data center to a disaster-tolerant multi-data-center design.

Amazon

APPLIED SCIENTIST INTERN

Seattle, WA

May 2021 - Aug. 2021

- Introduced a group-fairness notion for multi-armed bandits and developed a no-regret algorithm balancing fairness and performance.

University of Washington

VISITING RESEARCH SCIENTIST

Seattle, WA

Jan. 2021 - Apr. 2021

- Designed incentive-compatible, differentially private auction mechanisms for revenue maximization in repeated auctions.

Meta

RESEARCH INTERN, CORE DATA SCIENCE

Menlo Park, CA

May 2020 - Nov. 2020

- Developed optimal-spend-plan estimation and automatic bid-pacing algorithms for budget-constrained advertising campaigns.

Meta

MACHINE LEARNING ENGINEERING INTERN

Menlo Park, CA

May 2019 - July 2019

- Improved Dynamic Product Ads ranking and increased ad revenue while reducing model memory by 60 TB (86%).

Georgia Institute of Technology

GRADUATE RESEARCH ASSISTANT

Atlanta, GA

Aug. 2017 - Dec. 2022

- Developed robust learning algorithms with provable guarantees using online learning, algorithmic game theory, and differential privacy.

Education

Georgia Institute of Technology

PH.D. IN COMPUTER SCIENCE (ADVISORS: DR. JACOB ABERNETHY AND DR. JAMIE MORGENSTERN)

Atlanta, Georgia

2017 - 2022

Georgia Institute of Technology

M.S. IN COMPUTER SCIENCE (SPECIALIZATION IN MACHINE LEARNING, GPA: 4.0/4.0)

Atlanta, Georgia

2017 - 2022

Indian Institute of Technology, Kanpur

B.TECH IN COMPUTER SCIENCE AND ENGINEERING (GPA: 9.7/10)

Kanpur, India

2013 - 2017

Selected Recent Publications

- **[13] Semantic IDs for Recommender Systems at Snapchat: Use Cases, Technical Challenges, and Design Choices**
SIGIR 2026
C. M. Ju, T. Zhao, L. Neves, L. Collins, B. Kumar, et al.
- **[12] LLM-Based Generative Retrieval for Snapchat Content Recommendation**
2026 Preprint
L. Collins, J. Ren, D. Loveland, B. Kumar, et al.
- **[11] Exploiting ID-Text Complementarity via Ensembling for Sequential Recommendation**
RecSys 2026
L. Collins, B. Kumar, C. M. Ju, T. Zhao, D. Loveland, L. Neves, N. Shah
- **[10] Training-Free LLM-Based Recommendation with Post-LLM Item Refinement Using Collaborative Signals**
CIKM 2026
K. Kim, S. Kim, G. Lee, S. Kang, S. Kim, L. Collins, B. Kumar, et al.
- **[9] Sequential Data Augmentation for Generative Recommendation**
WSDM 2026
G. Lee, B. Kumar, C. M. Ju, T. Zhao, K. Shin, N. Shah, L. Collins
- **[8] CoSearch: Joint Training of Reasoning and Document Ranking via Reinforcement Learning for Agentic Search**
COLM 2026
H. Zeng, L. Collins, B. Kumar, N. Shah, H. Zamani
- **[7] Learning Universal User Representations Leveraging Cross-Domain User Intent at Snapchat**
SIGIR 2025
C. M. Ju, L. Neves, B. Kumar, L. Collins, T. Zhao, et al.
- **[6] Generative Recommendation with Semantic IDs: A Practitioner's Handbook**
CIKM 2025
C. M. Ju, L. Collins, L. Neves, B. Kumar, L. Y. Wang, T. Zhao, N. Shah
Best Paper Award; co-creator of GRID (700+ stars on GitHub)
- **[5] Revisiting self-attention for cross-domain sequential recommendation**
KDD 2025
C. M. Ju, L. Neves, B. Kumar, L. Collins, T. Zhao, et al.
- **[4] Private Mechanism Design via Quantile Estimation**
ICLR 2025
Y. Yang, T. Xiao, B. Kumar, J. Morgenstern
- **[3] ActiveHedge: Hedge meets Active Learning**
ICML 2022 (Spotlight)
B. Kumar, J. Abernethy, V. Saligrama
- **[2] Observation Free Attacks on Stochastic Bandits**
NeurIPS 2021
Y. Xu, B. Kumar, J. Abernethy
- **[1] Learning Auctions with Robust Incentive Guarantees**
NeurIPS 2019
B. Kumar, J. Abernethy, R. Cummings, J. Morgenstern, S. Taggart

Complete publication list: [Google Scholar](#).

Technical Skills

Programming & ML Python, C++, C; PyTorch, TorchRec, TensorFlow, verl
Distributed ML & Data Ray, vLLM, TensorRT-LLM; GCP (GKE, Vertex AI); SQL, Spark, Flink, Hive

Awards

2025 **Best Paper Award**, ACM CIKM
2019 **NeurIPS Travel Award**, Neural Information Processing Systems
2017 **Chair's Fellowship**, Georgia Institute of Technology
2014-16 **Academic Excellence Award (Dean's List)**, IIT Kanpur
2011 **KVPY Fellowship**, Government of India
2009 **NTSE Scholarship**, Government of India

Academic Service

Reviewer / Program Committee: ACL (2026); ICLR (2025, 2026); NeurIPS (2020, 2021, 2025); COLT (2020, 2021, 2023); SODA (2021, 2023); ICML (2022); JMLR (2021); ALT (2018, 2020).